Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Tri-Fusion
b. Team leader name: Vudumula Naga sai Rahul
c. Problem Statement: Sustainability & Social Impact

Team members: Mohammed Mudasser; Abubakr Siddique M

Brief about the idea

Namma Civic

Citizens verify the fix

Residents struggle to turn everyday civic problems into clear, actionable reports and to track whether a fix actually happened. Namma Civic connects accessible local reporting, community discussion, evidence-backed responses and citizen verification in one public record.

Photo + voice reporting **Real ward assignment** **Public outcome ledger**

Bengaluru prototype. Independent demonstration. No government endorsement.

Opportunities

Different by design

Independent evidence review

Officials claim completion. Eligible citizens confirm or dispute it. The ledger preserves the gap.

How it helps

One issue, one public history

Nearby reports join one canonical issue without deleting original reports. A ward queue makes responsibility visible.

Proposed USP

Bounded AI, auditable rules

Gemini observes and drafts. Code owns routing, priority, deadlines, transitions, quorum and every published metric.

List of features offered by the solution

Report copilot

Photo, voice or text; citizen confirms the draft.

Live civic record

Map, timeline, evidence, support, follow and Issue Pass.

Citizen verification

Eligible votes confirm the fix or reopen the issue.

Ward routing + duplicate review

Real polygons and distance checks before AI comparison.

Officer workflow

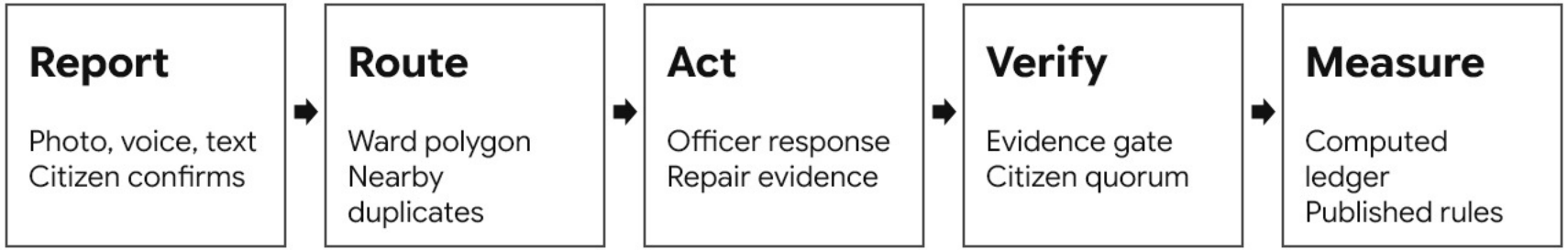
Acknowledge, start work, upload resolution evidence.

Outcome ledger

Claimed versus verified, SLA and reopen metrics with definitions.

Public Firebase Hosting device-only demonstration is accessible. Posts and interactions stay in each browser; connected backend features remain unavailable.

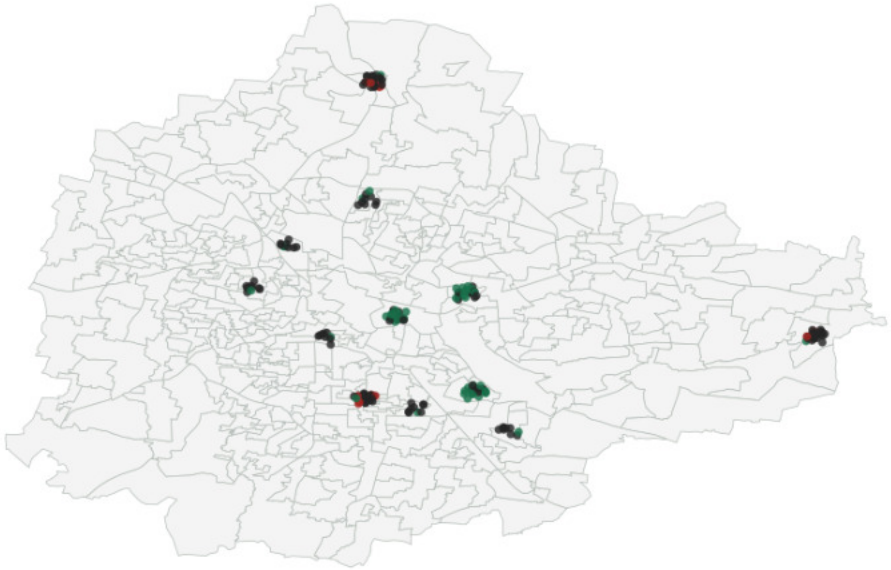
Process flow diagram or Use-case diagram



Wrong-location evidence is held for review

Only eligible citizens can verify or reopen. A written officer override stays visible. Silence never closes an issue.

Wireframes/Mock diagrams of the proposed solution (optional)



Map diagram from actual geometry

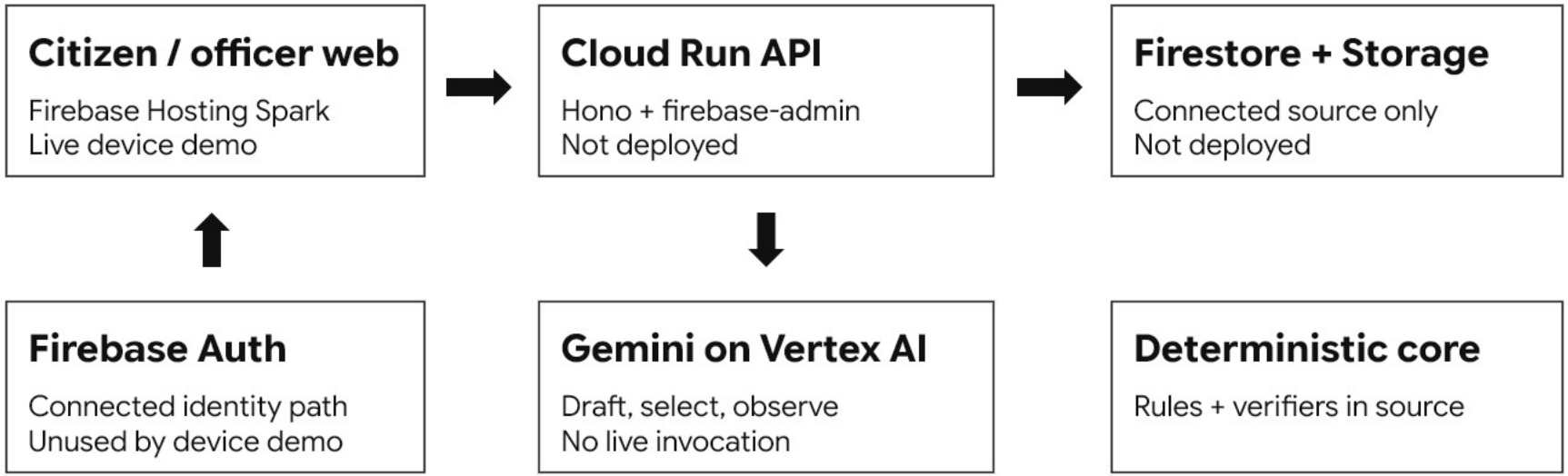
Illustrative ledger

100
Fixes claimed

50
Citizen verified

Scripted demo records
Not measured city performance
Not a product screenshot

Architecture diagram of the proposed solution



Live: Firebase Hosting Spark frontend only. Backend and Gemini are unavailable.
Target: API writes, schema checks, one repair retry, then a safe fallback.

Technologies to be used in the solution

Interface	React 19, TypeScript, Vite, Tailwind, Motion, MapLibre
Google services	Firebase Hosting Spark is live. Connected services are source only.
AI integration	Gemini on Vertex AI + verifiers in source. Live AI unavailable.
API and domain	Node.js, Hono, firebase-admin, shared deterministic rules
Privacy	API-only writes, EXIF removal, PII scrub, private citizen audio
Evidence	Original reports, append-only events, source attribution, demo labels

Estimated implementation cost (optional)

Firebase Hosting Spark frontend

Billing is unlinked. The static frontend uses Spark quotas.

Future Cloud Run	CPU, memory, requests and warm-instance time
Future Gemini	Input/output tokens and media processing
Future Firestore	Reads, writes, indexes and retention
Future Storage	Evidence size, downloads and retention

No paid backend is deployed or authorized. Spark quota limits apply. Future service costs need separate approval; no monthly cost measurement is claimed.

Snapshots of the prototype

Actual browser capture pending

The Firebase frontend is live. Verified product captures have not yet been selected for this deck. Illustrations are not presented as screenshots.

Required capture 1

Homepage and ward map
Include the visible demo-data label

Required capture 2

Issue evidence and officer workflow
Use the verified deployed build

Prototype Performance report/Benchmarking

Ward dataset	369 wards; 5 corporations; all representative points inside polygons
Seed integrity	200 demo histories replay; 1498 events; deterministic output hash
Media integrity	14 JPEGs without EXIF/XMP/IPTC; 17 sample fingerprints match
Code + Hosting	Typecheck and 9 HTTP tests pass. Live routes, media and layouts verified.

Firebase Hosting Spark is live. Billing is disabled.
Cloud Run, backend and Gemini remain unavailable.
No live AI quality or latency results are claimed.

Additional Details/Future Development (if any)

Connected backend	Firebase Hosting Spark is live. Cloud Run, backend and Gemini need separate approval, deployment and live verification.
First pilot	Work with one ward or resident group; obtain consent, define moderation and review disputed closures with humans.
Later development	Complaint-system integrations, notifications, privacy blurring, real official identity checks and larger-scale analytics.

Provide links to your:

GitHub public repository

https://github.com/Rahul21sai/LeadNet

Demo video (3-4 minutes)

[Owner: upload MP4 and paste public video URL]

Public prototype (Firebase device demo)

https://leadnet-blr.web.app/

All three links must open without account access during evaluation.

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Thank you

